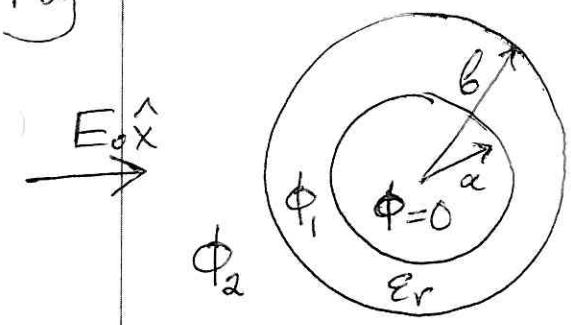
P2



Set $\phi=0$ on the conducting sphere.
General solution of Laplace eqn:

$$\phi(r,\theta) = \sum_\ell \left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}\right) P_\ell(\cos\theta)$$

Far from the sphere $\phi_2(r,\theta) = -E_0 r\cos\theta$
where θ = angle between $\hat{r}$ and $\hat{x}$.

$r<a$: $\phi = 0$

$a<r<b$: $\phi = \sum_\ell \left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}\right) P_\ell(\cos\theta)$

$r>b$: $\phi = -E_0 r\cos\theta + \sum_\ell \frac{C_\ell}{r^{\ell+1}} P_\ell(\cos\theta)$

Boundary conditions: $\phi(b-0) = \phi(b+0)$
$\phi(a-0) = \phi(a+0)$

also $D_\perp$ is continuous (no free charge)

$\Rightarrow \varepsilon_r \varepsilon_0 \frac{\partial\phi}{\partial r}$ at $r=b$, $r=a$ is continuous

1) $\phi(a-0) = 0 = \phi(a+0) = \sum_\ell \left(A_\ell a^\ell + \frac{B_\ell}{a^{\ell+1}}\right) P_\ell(\cos\theta)$

$\Rightarrow B_\ell = -A_\ell a^{2\ell+1}$ for $a \le r \le b$.

2) $\phi(b-0) = \sum_\ell A_\ell \left(b^\ell - \frac{a^{2\ell+1}}{b^{\ell+1}}\right) P_\ell(\cos\theta) = \phi(b+0) = -E_0 b\cos\theta$

$$+ \sum_\ell \frac{C_\ell}{b^{\ell+1}} P_\ell(\cos\theta)$$

3) $\varepsilon_r \left.\frac{\partial\phi}{\partial r}\right|_{r=b-0} = \left.\frac{\partial\phi}{\partial r}\right|_{r=b+0}$

$$\varepsilon_r \sum_\ell \left(A_\ell \ell b^{\ell-1} - B_\ell \frac{(\ell+1)}{b^{\ell+2}}\right) P_\ell(\cos\theta) = -E_0\cos\theta - \sum_\ell \frac{(\ell+1)C_\ell}{b^{\ell+2}} P_\ell$$

but from (1) $B_\ell = -A_\ell a^{2\ell+1}$

$$(3) \quad \varepsilon_r \sum_\ell A_\ell \left(\ell\, b^{\ell-1} + (\ell+1) \frac{a^{2\ell+1}}{b^{\ell+2}} \right) P_\ell(\cos\theta) =$$

$$= -E_0 \cos\theta - \sum \frac{(\ell+1) C_\ell}{b^{\ell+2}} P_\ell(\cos\theta)$$

Divide (2) by b:

$$(2') \quad \sum_\ell A_\ell \left(b^{\ell-1} - \frac{a^{2\ell+1}}{b^{\ell+2}} \right) P_\ell(\cos\theta) = -E_0 \cos\theta + \sum \frac{C_\ell}{b^{\ell+2}} P_\ell(\cos\theta)$$

Apply (2') and (3) to $\underline{\ell = 1}$ first:

$$A_1 \left(1 - \frac{a^3}{b^3} \right) \cancel{\cos\theta} = -E_0 \cancel{\cos\theta} + \frac{C_1}{b^3} \cancel{\cos\theta}$$

$$\varepsilon_r \left\{ A_1 \left(1 + 2 \frac{a^3}{b^3} \right) \right\} \cancel{\cos\theta} = -E_0 \cancel{\cos\theta} - \frac{2 C_1}{b^3} \cancel{\cos\theta}$$

Multiply first eqn. by 2 and add to second:

$$2 A_1 \left(1 - \frac{a^3}{b^3} \right) + \varepsilon_r A_1 \left(1 + 2 \frac{a^3}{b^3} \right) = -3 E_0$$

$$\Rightarrow A_1 = \frac{-3 E_0}{2 \left(1 - \frac{a^3}{b^3} \right) + \varepsilon_r \left(1 + 2 \frac{a^3}{b^3} \right)}$$

$$B_1 = -a^3 A_1$$

Now consider $\underline{\ell > 1}$:

$$A_\ell \left(b^{\ell-1} - \frac{a^{2\ell+1}}{b^{\ell+2}} \right) \cancel{P_\ell(\cos\theta)} = \frac{C_\ell}{b^{\ell+2}} \cancel{P_\ell(\cos\theta)}$$

$$\varepsilon_r A_\ell \left(\ell b^{\ell-1} + (\ell+1) \frac{a^{2\ell+1}}{b^{\ell+2}} \right) \cancel{P_\ell} = - \frac{(\ell+1) C_\ell}{b^{\ell+2}} \cancel{P_\ell}$$

Multiply first by $(\ell+1)$ and add to the second so that C_ℓ terms cancel in the right-hand-side $\Rightarrow A_\ell = 0$ for $\ell > 1$.

$\Rightarrow B_\ell = 0$ too.

$\Rightarrow$ for $a < r < b$

$$\phi(r,\theta) = A_1 \left(r - \frac{a^3}{r^2} \right) P_1(\cos\theta) =$$

$$= \frac{-3E_0}{2\left(1 - \frac{a^3}{b^3}\right) + \varepsilon_r \left(1 + 2\frac{a^3}{b^3}\right)} \left(r - \frac{a^3}{r^2} \right) \cos\theta$$

$$\vec{E}(r,\theta) = -\vec{\nabla}\phi = -\frac{\partial\phi}{\partial r}\hat{r} - \frac{1}{r}\frac{\partial\phi}{\partial\theta}\hat{\theta} =$$

$$= \frac{3E_0}{2\left(1 - \frac{a^3}{b^3}\right) + \varepsilon_r \left(1 + 2\frac{a^3}{b^3}\right)} \left\{ \left(1 + \frac{2a^3}{r^3}\right) \cos\theta \, \hat{r} \right.$$

$$\left. - \left(1 - \frac{a^3}{r^3}\right) \sin\theta \, \hat{\theta} \right\}.$$